LECTURE 1 — AMPUTATION OF EXTREMITIES

Definition

- **Amputation** = cutting the extremity (or part) **through the bone**. Cutting **through a joint = disarticulation**.

Indications (the "3 D's")

- **Dead/dying limb**: peripheral vascular disease, severe trauma, burns, frostbite.
- **Dangerous limb**: malignant tumours, lethal sepsis, **crush injury** → **crush syndrome**.
- **Damn nuisance limb** (keeping it is worse than losing it): pain, gross malformation, recurrent sepsis, severe loss of function.

Varieties

- **Provisional**: when primary healing unlikely — amputate as distal as cause allows, skin flaps loose over a pack, re-amputate later when favourable.
- **Definitive**: stump closed.
- **End-bearing** (weight through end): **scar NOT terminal**, bone end solid (cut near joint).
- **Non-end-bearing** (commonest): scar **can be terminal**; all upper limb + most lower limb. Examples: through-knee, **Syme's (above malleoli)**.

Sites of election

- Determined by: **prosthetic design demand** + **below the most distal palpable arterial pulsation**.
- **Too short stump** → slips out of prosthesis. **Too long stump** → painful, ulcerates, complicates joint incorporation. (Modern orthotists allow any level.)

Technique principles

- **Tourniquet** (unless vascular insufficiency).
- **Skin flaps**: combined length = **1.5× limb width** at amputation site. Equal flaps (upper limb + transfemoral); **long posterior flap for below-knee**.
- Muscle cut distal to bone level; **myoplasty** (muscles sutured over bone end).
- **Nerve: divide proximal to bone cut** (raw nerve end shouldn't bear weight).
- Bone: smooth ends, **bevel tibia anteriorly, fibula 3 cm shorter than tibia**. Remove tourniquet → haemostasis → suture skin without tension → suction drain → tight bandage.

Aftercare

- Evacuate haematoma early, **elastic bandage for shrinkage**, exercise muscles, keep joints mobile, advise prosthesis use.

Amputation levels

- **Upper limb**: interscapulothoracic, shoulder disarticulation, transhumeral, transradial, hand.
- **Lower limb**: hemipelvectomy, hip disarticulation, **transfemoral (≥12 cm)**, Gritti-Stokes (around knee), through-knee, **transtibial (14 cm)**, **Syme's (above malleoli)**, **Chopart (mid-tarsal)**, **Lisfranc (tarsometatarsal)**.

Complications of the stump

- **Early**: secondary haemorrhage, **skin flap breakdown** (ischaemia / suture tension), **gas gangrene**.

- **Late**:
- **Skin**: eczema, purulent lumps, ulceration, infected epidermoid cyst, **squamous cell carcinoma, verrucous hyperplasia**.
- **Muscle**: excessive muscle → unstable cushion.
- **Artery**: poor vascularity → cold, blue, ulcerating stump.
- **Nerve**: **painful neuroma** (tender nodule, **Tinel's sign**; overgrowth of fibrous tissue + nerve fibrils, NOT a tumour).
- **Phantom limb**: feeling limb still present; difficult to treat.
- **Joint**: stiffness/deformity (fixed flexion + abduction in above-knee; fixed flexion in below-knee).
- **Bone**: terminal **bone spur** (pain/infection), fracture (osteoporosis).

Ideal prosthesis

- Fits comfortably, functions well, looks presentable, fitted soon after operation.
- Lower limb weight transmitted through **greater trochanter, tibial tuberosity, patellar tendon, upper tibia, soft tissues**. Electrically powered available.

Amputation — high-yield

- **3 D's: Dead, Dangerous, Damn nuisance.**
- **Site = below most distal palpable pulse; skin flaps = 1.5× limb width; fibula 3cm shorter than tibia.**
- **Neuroma = Tinel's sign, not a tumour. Phantom limb hard to treat.**
- **Non-end-bearing = commonest (scar can be terminal); end-bearing = scar NOT terminal.**

LECTURE 2 — MSK INFECTION (Septic Arthritis, Osteomyelitis, Diabetic Foot)

Septic Arthritis (acute suppurative arthritis)

- **Definition**: joint infection by pus-forming organism.
- **Routes**: direct (penetrating wound, injection, arthroscopy); spread from adjacent bone abscess (infant — growth plate not a barrier); **intracapsular metaphysis (upper femur, upper humerus)** → **osteomyelitis easily causes septic arthritis**; metastatic blood spread.
- **Organisms**: **Staph aureus** (commonest); **H. influenzae (children 1–4 yrs)**; strep, E. coli, Proteus.
- **Clinical**:
- **Neonate**: emphasis on **septicaemia** (irritable, refuse feed, tachycardia, fever); joint warm/tender/resists movement; examine umbilical stump + IV site.
- **Child**: commonest **hip, knee, shoulder**; pseudoparesis, joint swelling, held in position of ease, warm, tender.
- **Adult**: superficial joints (knee, wrist, fingers, ankle); ask about **gonococcal infection + drug abuse**.
- **Investigations**: ↑WBC, ↑ESR, ↑CRP, **joint aspiration (pus → culture)**, blood culture, US, X-ray.
- **Treatment**: **antibiotics (flucloxacillin, 3rd-gen cephalosporin, fusidic acid)**, analgesics, IV fluids, **drainage if no response in 72h**, splintage.
- **Complications**: osteomyelitis, subluxation/dislocation, **epiphyseal plate damage (deformity/LLD)**, bony ankylosis, **septicaemia (lethal if untreated)**, anaemia.

Osteomyelitis

- **Definition**: infection of bone marrow by pyogenic organism; acute or chronic. Reaches bone by **direct spread** or **bloodstream**.
- **Acute haematogenous OM**: mainly **children** (+ adults with low immunity); trauma determines site.
- **Organisms**: **Staph aureus >70%**; strep pyogenes; group B strep (newborns); S. pneumoniae; **H. influenzae (1–4 yrs)**; Kingella kingae (post-URTI); **Salmonella in sickle cell**.
- **Progression**: inflammation → suppuration → bone necrosis → reactive new bone → resolution or **chronicity**.
- **Diagnosis**: CBC, ESR, CRP, blood + pus culture (periosteal abscess aspirate), X-ray, US.
- **DDx**: cellulitis, septic arthritis, necrotizing myositis, acute rheumatism, **sickle cell crisis**, Gaucher's, **Ewing sarcoma, osteosarcoma**, scurvy.
- **Treatment**: **empirical antibiotics first (don't wait for C&S)** then specific; surgical drainage if needed; splintage/rest; supportive.
- **Complications**: epiphyseal damage/altered growth, suppurative arthritis, metastatic infection, **pathological fracture, chronic osteomyelitis**, anaemia/toxaemia/death.

Diabetic Foot

- **Definition**: complication of long-standing DM — **infection/ulcer below the malleoli**. **30% have neuropathy/angiopathy; 25% develop diabetic foot.**

- **Peripheral vascular disease**: atherosclerosis of **medium vessels below knee**; pain + claudication, smooth cold skin, weak/absent pulses. **Superficial ulcers at toes, deep at pressure points (heel, 1st metatarsal plantar) — vascular ulcers painful + tender**.
- **Rocker-bottom deformity** (midfoot collapse, Charcot) = **diagnostic + disastrous** → ↑ulceration + infection risk.
- **Management** (multidisciplinary):
- **Prevention**: regular diabetic clinic, medication compliance, check vascular/neuro signs, foot care + footwear, skin hygiene.
- **Control hyperglycaemia** (oral agents ± insulin), **antibiotics by C&S**, daily saline dressing, **debridement (wash out biofilms)**.
- Surgery: **ray excision of infected toe**, transverse tibial bone transport, **Ilizarov frame** (off-load pressure), **total contact casts** (avoid prolonged bed rest).

MSK infection — high-yield

- **Septic arthritis: Staph aureus (H. influenzae in 1–4yr); aspirate joint; drain if no response in 72h; septicaemia = lethal.**
- **Osteomyelitis: Staph aureus >70%; Salmonella in sickle cell; empirical antibiotics first; children commonest.**
- **Diabetic foot: ulcer below malleoli; rocker-bottom (Charcot) deformity diagnostic; debridement + glycaemic control + C&S antibiotics; multidisciplinary.**
- **Neonatal septic arthritis = septicaemia picture (examine umbilical stump + IV site).**